TOTAL SETS IN QUANTUM FIELD THEORY

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It is shown for a certain class of sets in Minkowski space (called total ones) that the Haag algebra assigned to such a set coincides with the Haag algebra assigned to the whole Minkowski space. Some general criterion for "totality" is given (Theorem 1). Various types of total regions are constructed by means of this criterion.

Among the other examples, the open past light cone and the open isotropic tube are proved to be total sets.

For all sets showed in this paper to be total, the H. Araki's and H. J. Borchers' conditions do not give this result.

1. Introduction

Let us consider the algebraic formulation of quantum field theory introduced in [4]. For any open O of the Minkowski space M, $R(O)$ will mean the *v. Neumann algebra of operators* (acting in the Hilbert space H) associated with O.

Some explanation of the notation used below should be given. We draw a distinction between the Minkowski space M and its translation group denoted by $\vec{M}$. $\vec{M}$ is a four-dimensional real vector space endowed with a symmetric scalar product of the signature $(+, -, -, -)$, whereas M is an affine space associated with $\vec{M}$. Any vector $a \in \vec{M}$ defines a translation operation acting in M, it maps a point $x \in M$ onto a point denoted by $x+a$.

Let $U(\cdot)$ be the unitary representation of the translation group. Assume that $R(\cdot)$ and $U(\cdot)$ satisfy the following axioms:

I. For any family $(O_\alpha)_{\alpha \in \Lambda}$ of open subsets of M, $\mathscr{R}(\bigcup_\alpha O_\alpha)$ is the smallest v. Neumann algebra containing all algebras $\mathscr{R}(O_\alpha)$.

II. *Translation invariance*:

$$U(a)\mathscr{R}(O)U(a)^{-1} \subset \mathscr{R}(O+a)$$

for any open subset $O \subset M$ and any translation $a \in \vec{M}$.

III. *Spectral condition*: Let $dE(\cdot)$ be a spectral measure introduced by

$$U(a) = \int_{\vec{M}} e^{-iap} dE(p), \quad a \in \vec{M}. \tag{i}$$

Then the support Δ of this measure is contained in the closed future light cone:

$$\Delta \subset \{p \in \vec{M}: \ p^2 \geqslant 0 \ \text{ and } \ p_0 \geqslant 0\}.$$

IV. *Existence and uniqueness of the vacuum state*:

$$E(\{0\}) = |\Omega)(\Omega|, \tag{2}$$

where Ω is a normalized vector belonging to H and called the *vacuum vector*.

V. *Cluster property*: For any non-zero translation a and for any two vectors $\Psi, \Phi \in H$:

$$\lim_{\lambda \to \infty} (\Psi \mid U(\lambda a)\Phi) = (\Psi \mid \Omega)(\Omega \mid \Phi).$$

VI. *Cyclicity of the vacuum vector*: The set $\{A\Omega: A \in \mathcal{R}(M)\}$ is dense in H.

Remarks: 1. It is seen that Axiom IV can easily be proved by using the cluster property. On the other hand, the cluster property follows from Axiom IV, provided the representation $U(\cdot)$ is obtained by a restriction of a unitary representation of the Poincaré group.

2. The axioms above are sufficient to prove the Reeh–Schlieder theorem (see, for example, [2]).

An open set $O \subset M$ will be called *total* if and only if $\mathcal{R}(O) = \mathcal{R}(M)$. Simple examples of total sets are given in [3] and [1]. We shall give a few further examples of total sets. At first, we shall formulate an interesting sufficient condition for totality and later we shall show that this condition is very useful.

2. The main theorem

Let $\mathcal{W}(\cdot)$ be a complex function defined on $\vec{M}$. The function $\mathcal{W}(\cdot)$ will be called a *commutator function* if and only if

$$\mathcal{W}(a) = \int e^{-ipa} d\mu(p), \tag{3}$$

where $d\mu(\cdot)$ is a finite (not necessarily positive) measure with the support contained in $\Delta \cup (-\Delta)$. A commutator function $\mathcal{W}(\cdot)$ will be called *regular* if and only if

$$\lim_{\lambda \to \infty} \mathcal{W}(b + \lambda a) = 0 \tag{4}$$

for any $a, b \in \vec{M}$, provided $a \neq 0$.

Let $B(H)$ denote the set of all bounded operators acting in H; $A, B \in B(H)$; $B(a) \stackrel{\text{df}}{=} U(a)\, B\, U(a)^{-1}$ and

$$\mathcal{W}(a) \stackrel{\text{df}}{=} (\Omega \mid [A, B(a)]\Omega) \tag{5}$$

for any $a \in \vec{M}$. Then the function $\mathcal{W}(\cdot)$ introduced by the equation above is a regular

commutator function. Indeed, remembering that $U(a)\,\Omega=\Omega$ and using (1), one can check that equation (3) is satisfied by the measure

$$d\mu(p)\overset{\mathrm{df}}{=}(A^*\Omega\,|\,dE(p)\,B\Omega)-(B^*\Omega\,|\,dE(-p)\,A\Omega). \tag{6}$$

$\mathscr{W}(\cdot)$ is regular in virtue of the cluster property.

We say that a set $Q\subset\vec{M}$ *satisfies condition* (T) if and only if

$$(\mathscr{W}(a)=0 \text{ for any } a\in Q)\Rightarrow(\mathscr{W}(a)=0 \text{ for any } a\in\vec{M}) \tag{7}$$

for any regular commutator function $\mathscr{W}(\cdot)$.

THEOREM 1. *Let $\mathscr{U}$ be a region in M and $Q\subset\vec{M}$. Assume that Q satisfies condition* (T). *Then*

$$\mathscr{U}+Q=\{x+a:\ x\in\mathscr{U},\ a\in Q\}\subset M$$

is a total set. More exactly: $\mathscr{R}(\mathscr{U}+Q)=B(H)$.

Proof: Let $A\in\mathscr{R}(\mathscr{U}+Q)'$ and $B\in\mathscr{R}(\mathscr{U})$. Then $B(a)\in\mathscr{R}(\mathscr{U}+a)\subset\mathscr{R}(\mathscr{U}+Q)$ and $[A, B(a)]=0$ for any $a\in Q$. In virtue of (T) one can conclude that the regular commutator function (5) vanishes identically. Remembering that this function is the Fourier transform of measure (6) we have

$$(A^*\Omega\,|\,dE(p)\,B\Omega)=(B^*\Omega\,|\,dE(-p)\,A\Omega).$$

The supports of both sides of this equation are contained in Δ and $(-\Delta)$, respectively. The spectral conditions says that $\Delta\cap(-\Delta)=\{0\}$. Therefore

$$\operatorname{supp}(A^*\Omega\,|\,dE(\cdot)\,B\Omega)\subset\{0\}.$$

Taking into account (2) we obtain

$$(A^*\Omega\,|\,B\Omega)=\int\limits_{\vec{M}}(A^*\Omega\,|\,dE(p)\,B\Omega)=\int\limits_{\{0\}}(A^*\Omega\,|\,dE(p)\,B\Omega)=(A^*\Omega\,|\,\Omega)(\Omega\,|\,B\Omega),$$

and in a more convenient form:

$$(A^*\Omega\,|\,B\Omega)=(\bar{\lambda}\Omega\,|\,B\Omega),$$

where $\lambda\overset{\mathrm{df}}{=}(A^*\Omega\,|\,\Omega)$. This equation is satisfied by any operator $B\in\mathscr{R}(\mathscr{U})$, therefore

$$A^*\Omega=\bar{\lambda}\Omega$$

in virtue of the Reeh–Schlieder theorem. For any $C\in\mathscr{R}(\mathscr{U}+Q)$ we have

$$A^*C\Omega=CA^*\Omega=\bar{\lambda}C\Omega.$$

Using the Reeh–Schlieder theorem once more we obtain $A^*=\bar{\lambda}I$ and $A=\lambda I$. This way we have proved that any operator belonging to $\mathscr{R}(\mathscr{U}+Q)'$ is of the form λI, where $\lambda\in\boldsymbol{C}^1$. It means that $\mathscr{R}(\mathscr{U}+Q)'=\{\lambda I:\ \lambda\in\boldsymbol{C}^1\}$, and therefore

$$\mathscr{R}(\mathscr{U}+Q)=\{\lambda I:\ \lambda\in\boldsymbol{C}^1\}'=B(H),\qquad\text{q.e.d.}$$

3. Examples

We are going to give a few examples of sets Q satisfying condition (T). For convenience, we introduce an orthonormal coordinate system in $\vec{M}$. The coordinates of a vector $a \in \vec{M}$ are denoted by (a_α), where $\alpha = 0, 1, 2, 3$. The scalar product of two vectors $a, b \in \vec{M}$ is given by

$$ab = a_0 b_0 - \vec{a}\vec{b},$$

where $\vec{a} = (a_1, a_2, a_3)$, $\vec{b} = (b_1, b_2, b_3)$ and $\vec{a}\vec{b} = \sum_{k=1}^{3} a_k b_k$.

THEOREM 2. *Assume that the point* 0 *is separated from the rest of the energy-momentum spectrum. It means that*

$$E_0 \stackrel{\text{df}}{=} \operatorname{Inf}\{p_0 : p \in \Delta \text{ and } p \neq 0\} > 0. \tag{7}$$

Then the past light cone

$$V_- = \{a \in \vec{M} : a_0 < 0 \text{ and } a^2 > 0\}$$

satisfies condition (T).

Proof: Let $d\mu(\cdot)$ be a measure such that

$$\operatorname{supp} d\mu(\cdot) \subset \{p \in \vec{M} : p^2 \geqslant 0 \text{ and } |p_0| \geqslant E_0\} \cup \{0\}. \tag{8}$$

Assume that the commutator function

$$\mathcal{W}(a) = \int e^{-ipa} d\mu(p)$$

vanishes in the past light cone, i.e.:

$$\mathcal{W}(a) = 0 \tag{9}$$

for $a_0 < -\sqrt{\vec{a}^2}$. We have to prove that $\mathcal{W}(\cdot)$ vanishes identically.

For any $\vec{a} \in \boldsymbol{R}^3$ the measure $e^{i\vec{p}\vec{a}}\, d\mu(p)$ is finite and therefore it may be projected onto the p_0-axis. Denoting by $d\mu_{\vec{a}}(\cdot)$ the result of this projection we have

$$\mathcal{W}(a) = \int_{-\infty}^{+\infty} e^{-ip_0 a_0} d\mu_{\vec{a}}(p_0). \tag{10}$$

Equation (9) shows that the support of the Fourier transform of the measure $d\mu_{\vec{a}}(\cdot)$ is bounded below. It means that $d\mu_{\vec{a}}(\cdot)$ is the boundary value of a function holomorphic in the upper half plane. On the other hand, taking into account condition (8) one can see that $d\mu_{\vec{a}}(\cdot)$ vanishes in $]-E_0, 0[$ and $]0, E_0[$. It is known that the boundary value of a holomorphic function vanishes identically, provided it vanishes in an open set. Therefore, $d\mu_{\vec{a}}(p_0) = 0$ and according to (10): $\mathcal{W}(a) \equiv 0$, q.e.d.

Remark: It is known that any solution of the Klein–Gordon equation, with positive mass vanishing in the past light cone, vanishes identically [5]. Let us note that this result is, in fact, contained in Theorem 2.

COROLLARY 1. *Assume that condition* (7) *is satisfied. Then* $x_0 + V_-$ (*where* x_0 *is a fixed point of* M) *is a total set.*

Indeed, $x_0 + V_- = \mathcal{U} + Q$, where $\mathcal{U} = x_0 + V_-$, $Q = V_-$, and Theorem 1 can be applied.

COROLLARY 2. *Assume that condition* (7) *is satisfied. Let* l *be a time-like connected line which starts at the past infinity and ends at a point* $x_0 \in M$, *and let* O *be an open neighbourhood of* l. *Then* O *is a total set.*

Indeed, using Borchers result [3] one can conclude that $\mathcal{R}(O) \supset \mathcal{R}(x_0 + V_-)$.

For the general case, when condition (7) is violated, we have

THEOREM 3. *The set*

$$Q_3 \stackrel{\text{df}}{=} \{a \in \vec{M}:\ a_0 < -\sqrt{a_2^2 + a_3^2}\}$$

satisfies condition (T).

Proof: Assume that (3) is a regular commutator function vanishing on Q_3. For any continuous bounded function of one real variable we put

$$d\mu_\varphi(p) = \varphi(p_1)\, d\mu(p)$$

and

$$\mathcal{W}_\varphi(a) = \int e^{-ipa} d\mu_\varphi(p).$$

Let us notice that the set Q_3 is invariant under all translations parallel to the a_1-axis. This fact can be used in order to prove that $\mathcal{W}_\varphi(a) = 0$ for any $a \in Q_3$. In particular, this equation holds for any $a \in V_-$. (It is seen that $V_- \subset Q_3$.)

Assume that the function $\varphi(\cdot)$ vanishes on a neighbourhood $]-E_0, E_0[$ of the point 0. Then

$$\operatorname{supp} d\mu_\varphi(\cdot) \subset \{p \in \vec{M}:\ p^2 \geqslant 0 \text{ and } |p_1| \geqslant E_0\}.$$

Let us note that $|p_0| \geqslant E_0$ for any $p \in \vec{M}$ such that $p^2 \geqslant 0$ and $|p_1| \geqslant E_0$. Therefore

$$\operatorname{supp} d\mu_\varphi(\cdot) \subset \{p \in \vec{M}:\ p^2 \geqslant 0 \text{ and } |p_0| \geqslant E_0\}.$$

This condition looks like (8). It means that the argumentation given in the proof of Theorem 2 shows that $\mathcal{W}_\varphi(a) = 0$ for any $a \in \vec{M}$. This way we have proved that $\varphi(p_1)\, d\mu(p) = 0$ for any continuous bounded function $\varphi(\cdot)$ vanishing on a neighbourhood of the point 0. It means that

$$\operatorname{supp} d\mu(\cdot) \subset \{p \in \vec{M}:\ p_1 = 0\}.$$

Formula (3) shows now that the function $\mathcal{W}(\cdot)$ does not depend on a_1. Such a function is not regular (in our sense) unless it vanishes identically, q.e.d.

COROLLARY 3. *Let* s *be a space-like straight line in* M. *Then* $s + V_-$ *is a total set.*

Indeed, we may assume that s is parallel to the a_1-axis. Then $s + V_- = \mathcal{U} + Q_3$, where $\mathcal{U} = s + V_-$, and Theorem 1 can be applied.

COROLLARY 4. *Let* s *be a space-like straight line in* M *and* l *a time-like connected line which starts at the past infinity and ends at a point* $x_0 \in s$. *Then any open neighbourhood* O *of* $s \cup l$ *is a total set.*

Indeed, using Borchers results [3] one can conclude that $\mathscr{R}(O) \supset \mathscr{R}(s+V_-)$.

Let τ be a positive number,

$$R_\pm = \{(a_0, a_1) \in \boldsymbol{R}^2 : \ |a_0 \pm \tau| < \pm a_1\} \quad \text{and} \quad R = R_+ \cup R_- .$$

LEMMA. *Let $d\mu_m(\cdot)$ be a finite (not necessarily positive) measure defined on the hyperbola*

$$\Gamma_m = \{(p_0, p_1) \in \boldsymbol{R}^2 : \ p_0^2 - p_1^2 = m^2\}; \quad m > 0.$$

Assume that the function

$$\mathscr{W}_m(a_0, a_1) \overset{\text{df}}{=} \int_{\Gamma_m} e^{-i(p_0 a_0 - p_1 a_1)} d\mu_m(p_0, p_1)$$

vanishes in R. Then it vanishes identically.

Proof: Let us notice that

$$\boldsymbol{R}^1 - \{0\} \ni k \to \left(\frac{m}{2}\left(k + \frac{1}{k}\right), \frac{m}{2}\left(k - \frac{1}{k}\right)\right) \in \Gamma_m$$

is a homeomorphism. Denoting by $d\rho(\cdot)$ the reverse image of the measure $d\mu_m(\cdot)$ we have

$$\mathscr{W}_m(a_0, a_1) = \int_{\boldsymbol{R}^1 - \{0\}} e^{-\mathrm{im}\,[a_0(k+1/k) - a_1(k-1/k)]/2} d\rho(k)$$

and

$$\mathscr{W}_m(a_0, a_1) = \int_{\boldsymbol{R}^1 - \{0\}} e^{-\mathrm{im}\,(\xi k + \eta/k)} d\rho(k),$$

where

$$\xi \overset{\text{df}}{=} \tfrac{1}{2}(a_0 - a_1), \quad \eta \overset{\text{df}}{=} \tfrac{1}{2}(a_0 + a_1).$$

It can be checked that $(a_0, a_1) \in R$, and therefore

$$\int_{\boldsymbol{R}^1 - \{0\}} e^{-\mathrm{im}\,(\xi k + \eta/k)} d\rho(k) = 0 \tag{11}$$

for any (ξ, η) such that $|\eta| < \frac{1}{2}\tau$ and $|\xi| > \frac{1}{2}\tau$.

Assume that $|\eta| < \frac{1}{2}\tau$. Formula (11) says that the Fourier transform of the measure $e^{-\mathrm{im}\,(\eta/k)} d\rho(k)$ has a compact support. Therefore

$$e^{-\mathrm{im}\,(\eta/k)} d\rho(k) = r_\eta(k)\, dk,$$

where $r_\eta(\cdot)$ is an entire function. It is seen that

$$r_\eta(k) = e^{-\mathrm{im}\,(\eta/k)} r_0(k). \tag{12}$$

Assume that $r_0(\cdot)$ does not vanish identically. Then equation (12) implies that

$$e^{-\mathrm{im}\,(\eta/k)} = \frac{r_\eta(k)}{r_0(k)}. \tag{13}$$

On the other hand it is known that (13) is not a meromorphic function. This contradiction shows that $r_0(k) \equiv 0$, and therefore $d\rho(k) = r_0(k)dk = 0$, q.e.d.

Now we are able to prove

THEOREM 4. *The set*

$$Q_4=\{a\in\vec{M}\colon\ (a_0\,,\,a_1)\in R\}$$

satisfies condition (T).

Proof: Formula (3) means that a commutator function $\mathscr{W}(\cdot)$ is a superposition of plane waves with time-like (and isotropic) momentum vectors. It is convenient to rewrite this formula putting together the contribution introduced by the plane waves corresponding to a fixed value of $m=\sqrt{p_0^2-p_1^2}$, p_2 and p_3. This can be realized by using the Bourbaki theorem on measure desintegration. We obtain

$$\mathscr{W}(a)=\int_{\Delta} e^{i(p_2a_2+p_3a_3)}\mathscr{W}_{mp_2p_3}(a_0\,,\,a_1)\,d\sigma(m\,,\,p_2\,,\,p_3)\,, \tag{14}$$

where $d\sigma(\cdot)$ is a positive measure defined on the set $\Delta=\{(m,\,p_2\,,\,p_3)\colon\ m^2\geqslant p_2^2+p_3^2\}$ and for any $(m,\,p_2\,,\,p_3)\in\Delta$:

$$\mathscr{W}_{mp_2p_3}(a_0\,,\,a_1)=\int_{\Gamma_m} e^{-i(p_0a_0-p_1a_1)}d\mu_{mp_2p_3}(p_0\,p_1)\,, \tag{15}$$

where $d\mu_{mp_2p_3}$ is a finite measure defined on the hyperbola Γ_m. Let us note that function (15) satisfies the two-dimensional Klein–Gordon equation

$$\left[\left(\frac{\partial}{\partial a_0}\right)^2-\left(\frac{\partial}{\partial a_1}\right)^2+m^2\right]\mathscr{W}_{mp_2p_3}(a_0\,,\,a_1)=0\,.$$

Assume that the commutator function (14) vanishes in Q_4. Then, using the Jost–Lehman–Dyson representation technique [6] (see also [7]) one can prove that the functions $\mathscr{W}_{mp_2p_3}(\cdot)$ vanish in the region R. Now the lemma says that the only contribution to (14) is given by the point $(m,\,p_2\,,\,p_3)=(0,\,0,\,0)$, and therefore the function $\mathscr{W}(\cdot)$ does not depend on a_2 and a_3. Such a function is not regular (in our sense) unless it vanishes identically, q.e.d.

THEOREM 5. *For any positive number* τ *the set*

$$Q_5^{\tau}=\{a\in\vec{M}\colon\ |a_0-a_1|<\tau\}$$

satisfies condition (T).

Proof: Let (14) be a regular commutator function vanishing in Q_5^{τ}. Using the Jost–Lehman–Dyson representation technique one can prove that the functions $\mathscr{W}_{mp_2p_3}$ vanish in the region

$$\{(a_0\,,\,a_1)\colon\ |a_0-a_1|<\tau\}\,. \tag{16}$$

On the other hand it is easy to show that any solution of two dimensional Klein–Gordon equation (with positive mass) vanishes identically, provided it vanishes in (16). Therefore, the only contribution to (14) is given by the point $(m,\,p_2,\,p_3)=(0,\,0,\,0)$ and, in virtue of (4), we have $\mathscr{W}(a)\equiv 0$, q.e.d.

COROLLARY 5. *Let* $x_0\in M$ *and* $\tau>0$. *Then the set* $x_0+Q_5^{\tau}$ *is total.*

Indeed, $x_0+Q_5^{\tau}=O+Q_5^{\tau/2}$, where $O=x_0+Q_5^{\tau/2}$ and Theorem 1 can be applied.

COROLLARY 6. *Let O be an open subset of M. Then the isotropic tube $T \stackrel{\mathrm{df}}{=} O + \{a \in \vec{M} : a_0 = a_1, a_2 = a_3 = 0\}$ is a total set.*

Indeed, using Borchers result [3] one can check that

$$\mathscr{R}(T) = \mathscr{R}(O + \{a \in \vec{M} : a_0 = a_1\})$$

and Corollary 5 can be applied.

Remark: Let γ be a Poincaré transformation and $Q \subset \vec{M}$. It is seen that

$$(Q \text{ satisfies condition (T)}) \Leftrightarrow (\gamma Q \text{ satisfies condition (T)}).$$

This fact is very useful. It shows that the γ image of any above-mentioned total set is a total set.

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